NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

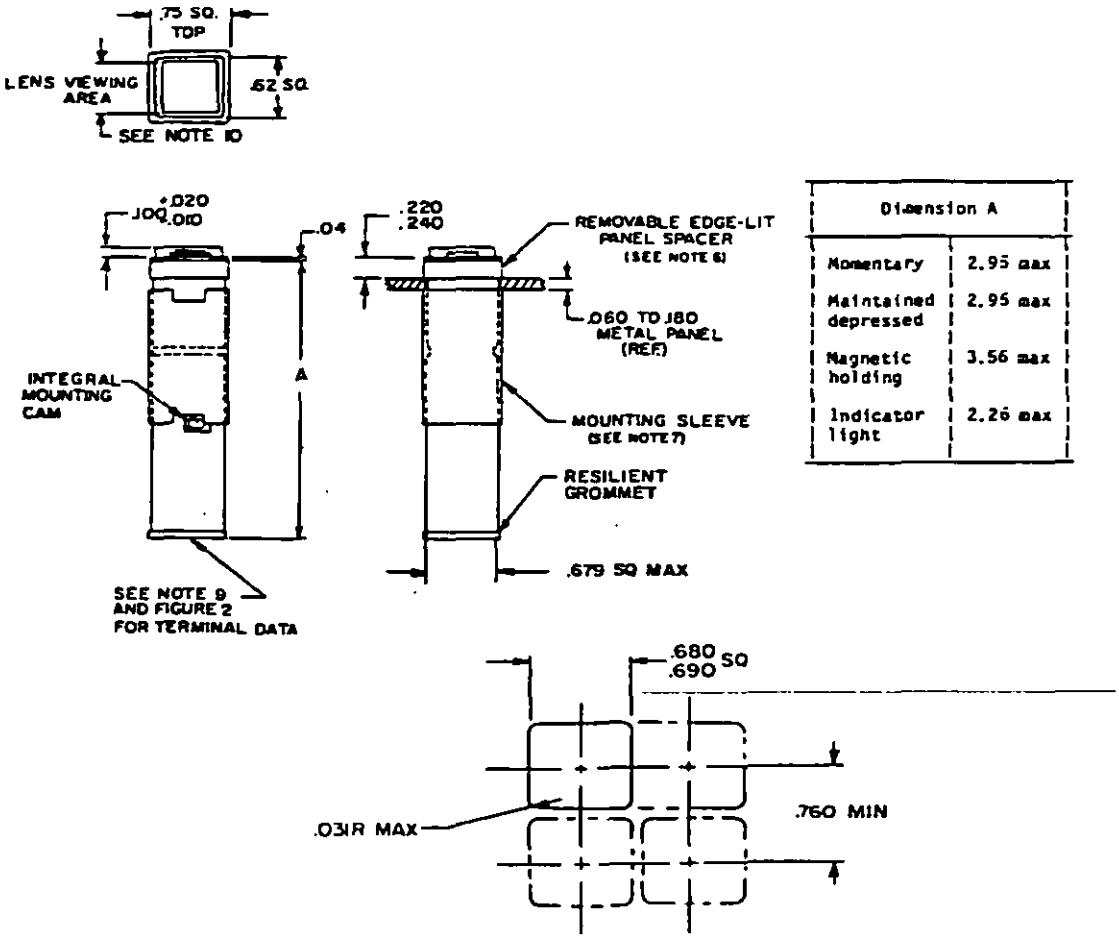
INCH-POUND
MIL-PRF-22885/77C
18 December 1987
SUPERSEDING
MIL-S-22885/77B
13 November 1978

PERFORMANCE SPECIFICATION SHEET

SWITCHES, PUSH BUTTON, ILLUMINATED, 4-LAMP, 0.75 SQUARE, 7.5 AMPERES, AND LOW LEVEL INTEGRATED WIRE TERMINATION, ELECTROMAGNETIC INTERFERENCE SHIELDED, SUNLIGHT READABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22885.

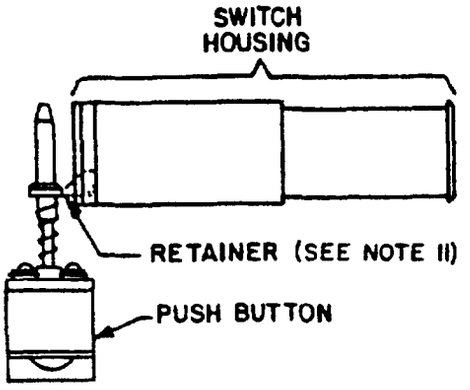


Dimension A	
Momentary	2.95 max
Maintained depressed	2.95 max
Magnetic holding	3.56 max
Indicator light	2.26 max

FIGURE 1. Switch.

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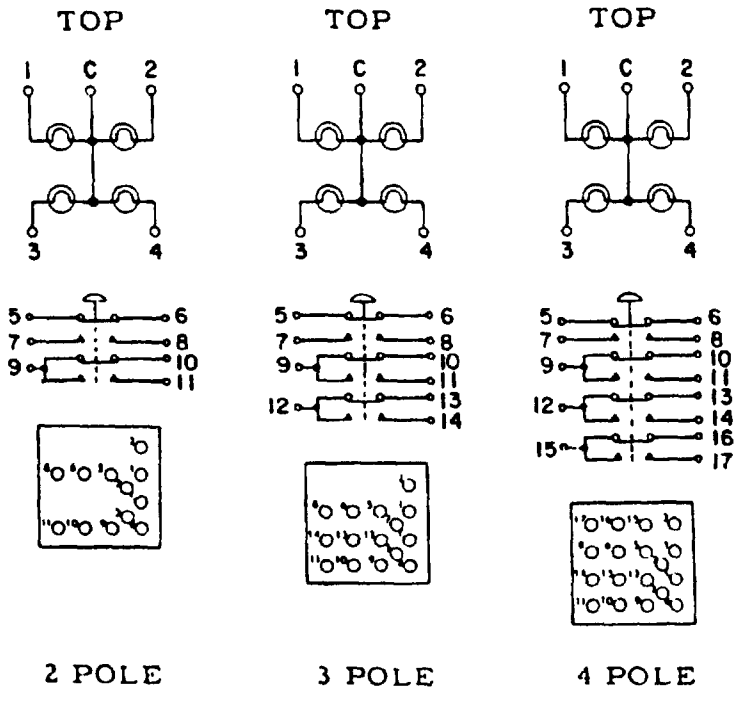


Inches	mm	Inches	mm	Inches	mm
.010	0.25	.180	4.57	.690	17.53
.020	0.51	.220	5.59	.75	19.0
.030	0.76	.240	6.10	.760	19.30
.031	0.79	.53	13.5	1.645	41.78
.04	1.0	.62	15.7	1.870	47.50
.060	1.52	.679	17.25		
.100	2.54	.680	17.27		

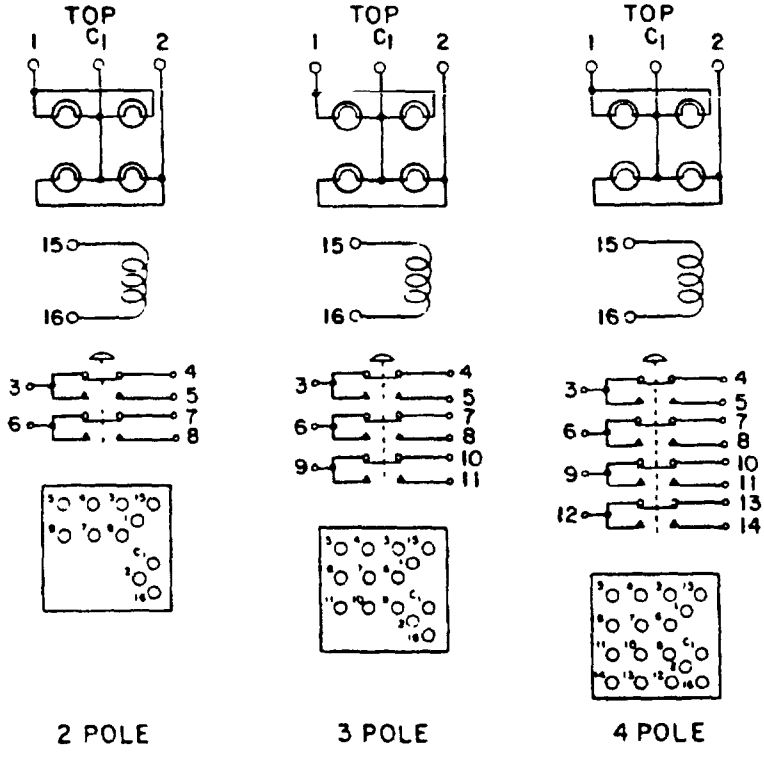
NOTES.
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
3. Metric equivalents are given for general information only.
4. Exact shape of switch optional provided dimensions specified are not exceeded.
5. A .194 (4.93 mm) thick spacer is supplied with each switch unit for MIL-P-7788 panel requirements.
6. Spacer places switch mounting flange flush with top of overlay panel. Spacer to be used only for edge-light panels, otherwise install behind panel ahead of sleeve.
7. Mounting sleeve shall not exceed .760 square and shall be reversible to accommodate edge-light panel requirements.
8. Pushbutton designed to prevent incorrect insertion into switch housing.
9. Terminations-Electrical connections shall be made by crimp, insertable-removable contacts and be environmentally sealed by a resilient grommet.
10. Available area for legend illumination is .53 square for display type W and .48 square for display type S.
11. Retained pushbutton shall be capable of having lamps replaced, but is restricted in complete removal from switch housing.

FIGURE 1. Switch - Continued.

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View 1 Momentary and maintained depressed alternate action.



View 2. Magnetic holding. Switch rotated 90° clockwise for legend plate style 2 Switch is available in legend plate styles 0, 1 and 2 only

NOTES
1. Lamp schematics indicate lamp positions as viewed from front of switch
2. Terminals shall be permanently marked on switch housing as shown

FIGURE 2 Schematic diagram.

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REQUIREMENTS:

Design and construction: See figure 1. Basic switches shall be MIL-S-8805/101 switches listed on QPL-8805.

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Vibration grade: 3 (10-2000 Hz).

Operation: A (Momentary action), D (Magnetically held), E (Position indicating, alternate action), H (Indicator light).

Illuminated color:

Display type W (White background) - See table I.

Display type S (Sunlight readable) - See table II.

TABLE I. Illuminated color - display type W.

Chromaticity coordinates 1/ 2200° kelvin		
Colors	X	Y
Green	.350 .350 .185 .265	SL 2/ .610 .610 .720
Yellow	.530 .544 .571 .582	.442 SL 2/ .402 SL 2/
Red	.680 .690 .715 .725	.300 .310 .265 .275
Blue	.195 .065 .095 .225	.305 .250 .150 .220
White	.484 Ref.	.415 Ref.
X is not less than .350. Y is not greater than .470. $Y - Y_o$ or $Y_o - Y$ is not numerically greater than 0.03X. (Y_o is the Y coordinate of the Planckian radiator for a given X).		

1/ The chromaticities of the diffusing lenses or colored filters expressed as X and Y coordinates on the CIE chromaticity diagram shall lie within areas bounded by the coordinates listed for each color, using a source illuminate of 2200 K.

2/ SL = spectrum locus (where intersected by other of coordinate pair).

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TABLE II. Illuminated color - display type S. 1/

Colors	X	Y
Red	.695	.285
	.710	SL 2/
	.655	.325
	.660	SL 2/
Green	.300	SL 2/
	.300	.600
	.380	.600
	.380	SL 2/
Yellow	.562	.415
	.570	SL 2/
	.596	.382
	.605	SL 2/
Blue	.230	.420
	.230	.350
	.320	.350
	.320	.420
White	.400	.420
	.460	.420
	.400	.380
	.460	.380

1/ The chromaticities of the colors are expressed as "X" and "Y" coordinates of the CIE chromaticity diagram and are within the limits bound by the coordinates listed for each color.
2/ SL - spectrum locus (where intersected by other coordinate pair).

Display type:

W (White until illuminated, then background appears in color).

S (Sunlight readable).

Material:

Housing: Corrosion resistant steel or aluminum alloy.

Termination: Terminals shall accept male pin contact M39029/1-101.

Insertion-extraction tools: M81969/14-02.

Sealing grommet: Silicone rubber.

Recommended wire diameter: .078 inch maximum.

Terminals shall meet the resistance to probe damage, maintenance aging, contact insertion and removal forces in accordance with the contact design, contact retention, and module design paragraphs of MIL-T-81714.

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Weight:

1.880 ounces maximum for momentary and alternation action and indicator light.

2.0 ounces maximum for magnetic hold.

Lamps:

Display type W pushbutton shall accommodate four lamps (T-1 3/4 midget flange base) except lens configuration 0 (see figure 3) shall accommodate only two lamps.

Display type S (sunlight readable) illumination is by four 5 volt, T-1, .15 MSCP lamps. Other T-1 lamps are available, with different illumination results. Lamps shall be acquired separately.

Operating characteristics:

Actuation force: 3.5 pounds ±1.5 pounds.

Actuation travel: Actuation occurs when the face of the pushbutton is below the mounting flange.

Lens extraction force: 1 1/2 to 4 pounds.

Holding coil. 0.04 ±.01 amperes at 28 V dc Will hold in at 18 V dc and drop out at 14 V dc.

Shock. 75 G sawtooth (MIL-STD-202, method 213, test condition H).

Moisture resistance. Following step 6 of the final cycle of the moisture resistance test, while switches are still wet, the insulation resistance measurement shall be not less than 5 megohms.

Electrical endurance: See tables III and IV for electrical load conditions. For magnetically held switches, one cycle shall consist of a mechanical actuation, an electrical hold, and an electrical release.

TABLE III. Electrical ratings - silver contacts. 1/ 2/

Load	Sea level		50,000 feet 28 V dc
	28 V dc	115 V ac, 60 Hz	
	(Amperes)	(Amperes)	(Amperes)
Resistive	7.5	7.5	4
Inductive 3/	5.0	5.0	2.5
Lamp 4/	1.0	1.0	1.0

1/ Minimun conditions are dependent on use and atmospheric conditions.
2/ For multiple pole switching of maximum rated loads, derate maximum load rating 30%.
3/ L/R - .026.
4/ Inrush, 10:1 for 20 milliseconds maximum.

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TABLE IV. Electrical ratings - gold plated contacts.

Load	28 V dc Sea level	28 V dc 50,000 feet
	Amperes	Amperes
Resistive	1	1
Inductive	.5	.5

Low level life: Applicable to gold contact switches.

EMI/RFI shielding: Shielding efficiency test applicable, except the shielding attenuation shall be not less than 47 dB at 100 MHz, 39 dB at 400 MHz and 33 dB at 1,000 MHz.

Luminance:

Display type W: See table V.

Display type S: Sunlight readability contrast requirements apply.

TABLE V. Luminance for display type W.

Color	Color code	Brightness 1/ (foot-lamberts, minimum average)
Blue	B	8.0
Green	G	50.0
Red	R	30.0
White	W	180.0
Yellow	Y	300.0

1/ When illuminated by two MS25237-387 lamps of 0.30 ±0.02 mean spherical candlepower at 28 volts dc, readings are for lens configuration 0. Luminance is without EMI screen. Brightness is reduced by approximately 50% when EMI screen is specified.

Sunlight readability: Applicable to display type S switches, see table VII. Units shall be tested for $\theta_1 = 45^\circ$ and $\theta_2 = 0^\circ$ only. This procedure does not test readability under specularly reflective (glare) condition. The average contrast ratio of each legend character to the background shall be 0.6 minimum in the lighted legend condition. In the unlighted legend condition, the average contrast ratio of each legend character to the background shall have an absolute value less than or equal to 0.1.

Qualfication inspection: All part numbers will be approved for inclusion in the qualified products list upon successful completion of testing specified in table VI. Table VI is based on use of qualified MIL-S-8805/101 basic switches in the assembly.

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TABLE VI. Qualification inspection - group submission.

Inspection 1/	Sample
Group I	All samples
Group II	M22885/77-632RG 3/ (4 samples) M22885/77-065R (4 samples)
Group III	M22885/77-018RG (2 samples)
Group VI 2/	M22885/77-018RG (2 samples) M22885/77-453RG 3/ (2 samples) M22885/77-065R (2 samples)
Group VII 4/	M22885/77-632RG 3/ (2 samples)
Group VIII	Type S display: 2 pushbuttons of each color (10 samples), of lens configuration 0, and with EMI/RFI shielding. 3/ 5/ Type W display: 2 pushbuttons of each color (10 samples), of lens configuration 0, and without EMI/RFI shielding. 5/

1/ Inspection groups in accordance with qualification inspection table of MIL-S-22885.

2/ Electrical endurance:
Silver contacts - test resistive load, ac, only.
Gold contacts - test sea level resistive, 28 V dc only.
Lamps used during electrical endurance test shall be MS25237-387 for type W display and MS24515-718 for type S display.

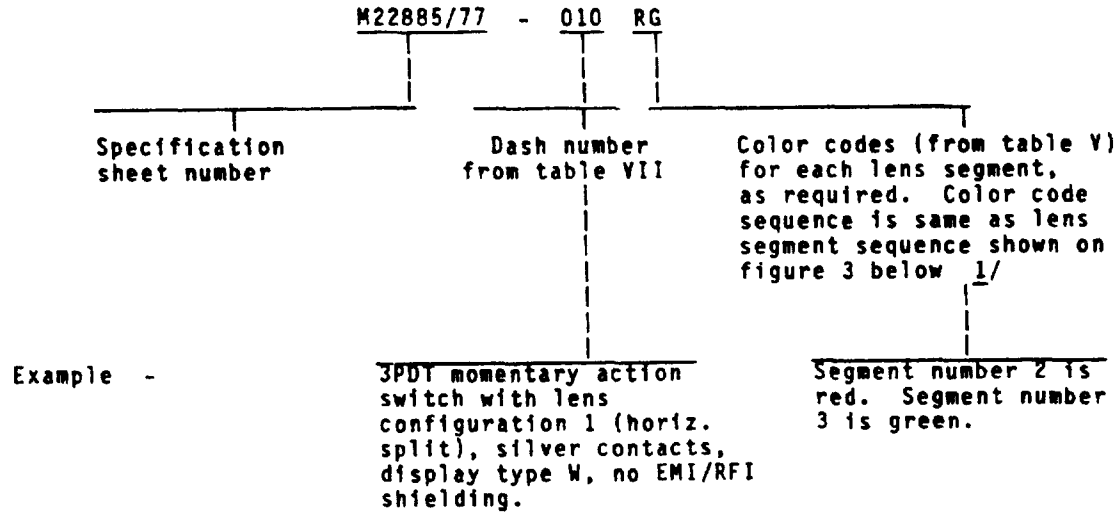
3/ Legend shall be in accordance with MIL-S-22885 sunlight readability test paragraph.

4/ Low level life test not required.

5/ Pushbuttons shall be assembled to any representative switch body.

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Part numbers: Part numbers are assigned as illustrated below. Part numbers do not include legend description. Acquire switches with legends as category II items in accordance with MIL-S-22885.



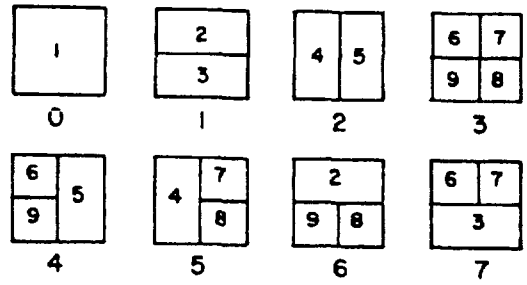


FIGURE 3 Lens configuration and segment numbers.

1/ Part number constructed as shown replace and are interchangeable with previous part numbers which included segment numbers in addition to color codes (example: M22885/77-010-2R3G).

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TABLE VII. Dash numbers.

Dash number		Operation	Circuit	Lens configuration (see figure 3)	Contact material	Display type
Non EMI shielded	EMI shielded					
001	217	Momentary	2PDT	0	Silver	W
002	218	"	"	1	"	"
003	219	"	"	2	"	"
004	220	"	"	3	"	"
005	221	"	"	4	"	"
006	222	"	"	5	"	"
007	223	"	"	6	"	"
008	224	"	"	7	"	"
009	225	Momentary	3PDT	0	Silver	W
010	226	"	"	1	"	"
011	227	"	"	2	"	"
012	228	"	"	3	"	"
013	229	"	"	4	"	"
014	230	"	"	5	"	"
015	231	"	"	6	"	"
016	232	"	"	7	"	"
017	233	Momentary	4PDT	0	Silver	W
018	234	"	"	1	"	"
019	235	"	"	2	"	"
020	236	"	"	3	"	"
021	237	"	"	4	"	"
022	238	"	"	5	"	"
023	239	"	"	6	"	"
024	240	"	"	7	"	"
025	241	E 1/	2PDT	0	Silver	W
026	242	"	"	1	"	"
027	243	"	"	2	"	"
028	244	"	"	3	"	"
029	245	"	"	4	"	"
030	246	"	"	5	"	"
031	247	"	"	6	"	"
032	248	"	"	7	"	"
033	249	E 1/	3PDT	0	Silver	W
034	250	"	"	1	"	"
035	251	"	"	2	"	"
036	252	"	"	3	"	"
037	253	"	"	4	"	"
038	254	"	"	5	"	"
039	255	"	"	6	"	"
040	256	"	"	7	"	"
041	257	E 1/	4PDT	0	Silver	W
042	258	"	"	1	"	"
043	259	"	"	2	"	"
044	260	"	"	3	"	"
045	261	"	"	4	"	"
046	262	"	"	5	"	"
047	263	"	"	6	"	"
048	264	"	"	7	"	"

See footnote at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number		Operation	Circuit	Lens configuration (see figure 3)	Contact material	Display type
Non EMI shielded	EMI shielded					
049	265	Mag. hold	2PDT	0	Silver	W
050	266	"	"	1	"	"
051	267	"	"	2	"	"
057	268	Mag. hold	3PDT	0	Silver	W
058	269	"	"	1	"	"
059	270	"	"	2	"	"
065	271	Mag. hold	4PDT	0	Silver	W
066	272	"	"	1	"	"
067	273	"	"	2	"	"
274	282	Indicator	Indicator	0	N/A	W
275	283	"	"	1	"	"
276	284	"	"	2	"	"
277	285	"	"	3	"	"
278	286	"	"	4	"	"
279	287	"	"	5	"	"
280	288	"	"	6	"	"
281	289	"	"	7	"	"
290	469	Momentary	2PDT	0	Gold	W
291	470	"	"	1	"	"
292	471	"	"	2	"	"
293	472	"	"	3	"	"
294	473	"	"	4	"	"
295	474	"	"	5	"	"
296	475	"	"	6	"	"
297	476	"	"	7	"	"
298	477	Momentary	3PDT	0	Gold	W
299	478	"	"	1	"	"
300	479	"	"	2	"	"
301	480	"	"	3	"	"
302	481	"	"	4	"	"
303	482	"	"	5	"	"
304	483	"	"	6	"	"
305	484	"	"	7	"	"
306	485	Momentary	4PDT	0	Gold	W
307	486	"	"	1	"	"
308	487	"	"	2	"	"
309	488	"	"	3	"	"
310	489	"	"	4	"	"
311	490	"	"	5	"	"
312	491	"	"	6	"	"
313	492	"	"	7	"	"

See footnote at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number		Operation	Circuit	Lens configuration (see figure 3)	Contact material	Display type
Non EMI shielded	EMI shielded					
314	493	E 1/	2PDT	0	Gold	W
315	494	"	"	1	"	"
316	495	"	"	2	"	"
317	496	"	"	3	"	"
318	497	"	"	4	"	"
319	498	"	"	5	"	"
320	499	"	"	6	"	"
321	500	"	"	7	"	"
322	501	E 1/	3PDT	0	Gold	W
323	502	"	"	1	"	"
324	503	"	"	2	"	"
325	504	"	"	3	"	"
326	505	"	"	4	"	"
327	506	"	"	5	"	"
328	507	"	"	6	"	"
329	508	"	"	7	"	"
330	509	E 1/	4PDT	0	Gold	W
331	510	"	"	1	"	"
332	511	"	"	2	"	"
333	512	"	"	3	"	"
334	513	"	"	4	"	"
335	514	"	"	5	"	"
336	515	"	"	6	"	"
337	516	"	"	7	"	"
338	517	Mag. hold	2PDT	0	Gold	W
339	518	"	"	1	"	"
340	519	"	"	2	"	"
341	520	Mag. hold	3PDT	0	Gold	W
342	521	"	"	1	"	"
343	522	"	"	2	"	"
344	523	Mag. hold	4PDT	0	Gold	W
345	524	"	"	1	"	"
346	525	"	"	2	"	"
347	526	Momentary	2PDT	0	Silver	S
348	527	"	"	1	"	"
349	528	"	"	2	"	"
350	529	"	"	3	"	"
351	530	"	"	4	"	"
352	531	"	"	5	"	"
353	532	"	"	6	"	"
354	533	"	"	7	"	"

See footnote at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number		Operation	Circuit	Lens configuration (see figure 3)	Contact material	Display type
Non EMI shielded	EMI shielded					
355	534	Momentary	3PDT	0	Silver	S
356	535	"	"	1	"	"
357	536	"	"	2	"	"
358	537	"	"	3	"	"
359	538	"	"	4	"	"
360	539	"	"	5	"	"
361	540	"	"	6	"	"
362	541	"	"	7	"	"
363	542	Momentary	4PDT	0	Silver	S
364	543	"	"	1	"	"
365	544	"	"	2	"	"
366	545	"	"	3	"	"
367	546	"	"	4	"	"
368	547	"	"	5	"	"
369	548	"	"	6	"	"
370	549	"	"	7	"	"
371	550	E 1/	2PDT	0	Silver	S
372	551	"	"	1	"	"
373	552	"	"	2	"	"
374	553	"	"	3	"	"
375	554	"	"	4	"	"
376	555	"	"	5	"	"
377	556	"	"	6	"	"
378	557	"	"	7	"	"
379	558	E 1/	3PDT	0	Silver	S
380	559	"	"	1	"	"
381	560	"	"	2	"	"
382	561	"	"	3	"	"
383	562	"	"	4	"	"
384	563	"	"	5	"	"
385	564	"	"	6	"	"
386	565	"	"	7	"	"
387	566	E 1/	4PDT	0	Silver	S
388	567	"	"	1	"	"
389	568	"	"	2	"	"
390	569	"	"	3	"	"
391	570	"	"	4	"	"
392	571	"	"	5	"	"
393	572	"	"	6	"	"
394	573	"	"	7	"	"
395	574	Mag. hold	2PDT	0	Silver	S
396	575	"	"	1	"	"
397	576	"	"	2	"	"
398	577	Mag. hold	3PDT	0	Silver	S
399	578	"	"	1	"	"
400	579	"	"	2	"	"

See footnote at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number		Operation	Circuit	Lens configuration (see figure 3)	Contact material	Display type
Non EMI shielded	EMI shielded					
401	580	Mag. hold	4PDT	0	Silver	S
402	581	"	"	1	"	"
403	582	"	"	2	"	"
404	583	Indicator	Indicator	0	N/A	S
405	584	"	"	1	"	"
406	585	"	"	2	"	"
407	586	"	"	3	"	"
408	587	"	"	4	"	"
409	588	"	"	5	"	"
410	589	"	"	6	"	"
411	590	"	"	7	"	"
412	591	Momentary	2PDT	0	Gold	S
413	592	"	"	1	"	"
414	593	"	"	2	"	"
415	594	"	"	3	"	"
416	595	"	"	4	"	"
417	596	"	"	5	"	"
418	597	"	"	6	"	"
419	598	"	"	7	"	"
420	599	Momentary	3PDT	0	Gold	S
421	600	"	"	1	"	"
422	601	"	"	2	"	"
423	602	"	"	3	"	"
424	603	"	"	4	"	"
425	604	"	"	5	"	"
426	605	"	"	6	"	"
427	606	"	"	7	"	"
428	607	Momentary	4PDT	0	Gold	S
429	608	"	"	1	"	"
430	609	"	"	2	"	"
431	610	"	"	3	"	"
432	611	"	"	4	"	"
433	612	"	"	5	"	"
434	613	"	"	6	"	"
435	614	"	"	7	"	"
436	615	E 1/	2PDT	0	Gold	S
437	616	"	"	1	"	"
438	617	"	"	2	"	"
439	618	"	"	3	"	"
440	619	"	"	4	"	"
441	620	"	"	5	"	"
442	621	"	"	6	"	"
443	622	"	"	7	"	"

See footnote at end of table.

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TABLE VII. Dash numbers - Continued.

Dash number		Operation	Circuit	Lens configuration (see figure 3)	Contact material	Display type
Non EMI shielded	EMI shielded					
444	623	E 1/	3PDT	0	Gold	S
445	624	"	"	1	"	"
446	625	"	"	2	"	"
447	626	"	"	3	"	"
448	627	"	"	4	"	"
449	628	"	"	5	"	"
450	629	"	"	6	"	"
451	630	"	"	7	"	"
452	631	E 1/	4PDT	0	Gold	S
453	632	"	"	1	"	"
454	633	"	"	2	"	"
455	634	"	"	3	"	"
456	635	"	"	4	"	"
457	636	"	"	5	"	"
458	637	"	"	6	"	"
459	638	"	"	7	"	"
460	639	Mag. hold	2PDT	0	Gold	S
461	640	"	"	1	"	"
462	641	"	"	2	"	"
463	642	Mag. hold	3PDT	0	Gold	S
464	643	"	"	1	"	"
465	644	"	"	2	"	"
466	645	Mag. hold	4PDT	0	Gold	S
467	646	"	"	1	"	"
468	647	"	"	2	"	"

1/ Maintained depressed alternate action.

NOTE: 800 and 900 series dash numbers, for switch housing only and pushbutton only, have been canceled without replacement.

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NOTE: Revision letter are not use to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Navy - AS, OS
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - MC, SH
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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